

POTPRESS · REQUIREMENTS

PotPress requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/potpress		
Source	docs/03-requirements.md @ 1b4da66		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

PotPress requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with a filter factory, and will be checked by calculation at TRL 3 and revised after co-design sessions (see PPR-PRB-001). One requirement is not met by the concept (R10, cost) and three are at risk (R2, R5 and R7); see Table 2.

The **reference filter** used throughout is the common flowerpot form: inner rim diameter 280 mm, inner depth 240 mm, wall 15 mm, flat rim about 345 mm across, about 12 L to the brim and about 10 L working volume (dimensions are estimates based on the 280 by 250 mm form reported by [Potters for Peace](#)).

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Form the reference filter	Inner rim 280 mm, inner depth 240 mm, wall 15 mm, flat rim; geometry set by parameters in one source file	Model check; later measurement of pressed pots
R2	Even wall	Wall thickness within ± 1 mm around the circumference and from pot to pot; molds self-center to within 0.5 mm	Tolerance stack and guide design; later sectioned pots
R3	Pressing force and frame strength	20 t (196 kN) rated jack; frame, pins and platen carry 1.5 times the jack rating (294 kN) without yield and with deflection under 1 mm at the mold	Beam and pin calculation; later proof load
R4	Throughput	Press cycle of 6 min or less with two operators, so 50 or more pots per 6 h of pressing	Cycle breakdown; later timed trial
R5	Mold opening	Male mold lifts at least 30 mm clear of the female rim so the mold carriage slides out for loading and demolding; pressing stroke within one jack stroke	Kinematic layout
R6	Flow-rate QC	Rack tests 4 soaked filters at once; T-gauge reads flow over 1 h to 0.1 L/h; repeatability ± 0.1 L/h; acceptance band set per factory (default 1.0 to 2.5 L/h in the first hour, proposed)	Gauge calibration by volume; later repeat tests
R7	Molds made locally	Molds cast from 3D-printed patterns printed on a 250 x 250 mm class printer; finishing with hand tools and a drill press; no lathe over 300 mm swing	Pattern split and finishing review with a foundry
R8	Frame built locally	Standard steel channel and plate; stick welding and drilling only; heaviest single part 40 kg or less for two-person handling	Part list and mass estimate
R9	Safe operation	Fixed guards on the sides and back, interlocked front gate; jack pump and release outside the guard; load pins must be in place before pressing	Hazard review; later guard check
R10	Affordable	Press plus QC rack \$600 or less in parts	Priced BOM (bom/ bom.csv)
R11	Footprint	Press within 1.0 x 0.7 m floor area and 2.0 m height; QC rack within 0.8 x 0.5 m	Model check
R12	Product-safe materials	Faces that touch clay or test water are aluminum, food-grade polyethylene or stainless steel; no lead-based paint or oiled release agents on mold faces	Material list review

Table 2. Concept status against each requirement (estimates from PPR-PRC-001).

ID	Concept estimate	Status
R1	Geometry parameters in <code>cad/src/concept_media.py</code>	Met on paper
R2	Depends on guide sleeve clearance and mold alignment, not yet designed	At risk , unverified
R3	Top beam of two UPN 160 channels (about 232 cm ³): about 127 MPa at the 196 kN rating and about 190 MPa at 294 kN, below 275 MPa yield	Met on paper; deflection and pins not checked
R4	About 5 min per cycle	Met on paper, thin margin
R5	Needs about 270 mm of relative travel; a standard bottle jack gives about 150 mm, so the male mold is lifted separately by a crank	At risk until the lift and jack are chosen
R6	Printed T-gauge, 4 stations	Met on paper
R7	Female pattern about 410 mm across must be printed in segments; finish of the cast cavity without a lathe is unproven	At risk
R8	Heaviest part is the top beam at about 35 kg (estimate)	Met on paper
R9	Guards and gate in the BOM, not modeled	Met on paper
R10	About \$716 in parts (press about \$645, QC rack about \$71)	Not met , about 19 % over
R11	Press about 920 x 640 x 1,655 mm; rack 800 x 440 mm	Met
R12	Aluminum molds, polyethylene liners and buckets	Met on paper

■ Assumptions

- Pressing force needed for a well-consolidated wall is not published in the sources found; 5 to 10 t is assumed as the working range and the 20 t jack rating is used as the design load (estimate, to be confirmed with a partner factory).
- Mix and mass per pot follow the factory recipes reported by [Rayner \(2009\)](#): about 8 kg of mix per pot.
- The default flow acceptance band of 1.0 to 2.5 L/h follows Nicaraguan practice and the Potters for Peace range of 1.5 to 2.5 L/h; each factory sets its own band.
- Allowable stress in structural steel is taken as 165 MPa (S275 with a factor of about 1.67).